



Stag beetle

Lucanus cervus

This large forest insect is famed for the enormous mandibles of the male, which are used in jousting fights. They resemble the antlers used by male deer in battles of strength over potential mates, hence the name stag beetle. Females are smaller than males, and although their mandibles are more discreet, they can grip more strongly than the males. Like all beetles, both sexes have a hard, armored shell for protection.

Fueled by fat reserves

Adult stag beetles do not feed. Instead, they rely on their fat reserves built up while they were larvae living underground. They will, however, occasionally sip tree sap or juices from decomposing fruit using their smaller hairy mouthparts. Otherwise, the adults devote themselves to mating.

The males dig their way out of the ground in May or June, a week or so before the females. They establish a mating territory and use their “antlers” to fight off any late arrivals hoping to muscle in. The females walk from territory to territory, during which time they mate with several males. Adult life rarely lasts more than three months. The last thing a female does before she dies is find a suitable piece of rotting wood—usually decaying

tree stumps or roots—where she lays about 20 eggs. Sometimes, females return to the site where they were larvae to lay their eggs.

Growing up underground

In total, a stag beetle lives for about six years. Most of this time is spent as a larva, eating wood rotting underground. The eggs hatch in August, and the tiny orange-headed grubs begin a five-year feeding session. It takes this long for a grub to reach full size and build up the crucial fat supply for the adult phase. When ready, the grub builds a cocoon chamber out of chewed wood fibers and turns into a pupa. In this form, it remains immobile for at least two months as its larval body is broken down and rebuilt as an adult. The pupa is protected by a hard case and the sex of the beetle can be identified—the large mandibles of a male are already visible. Pupation occurs in fall, but once emerged, the adult stays underground for the ensuing winter, readying itself to emerge the following summer.

The stag beetle population across the world is rapidly declining for many reasons, including changes in forest management that has led to the removal of dead timber. Increased urbanization is also a threat.

Stag beetles spend almost **their entire lives underground** as larvae

- ↔ 3 in (7.5 cm)
- ⊗ Near threatened
- 🌳 Dead wood, sap, fruit juices
- 🏠 



Europe, Asia



▷ JOUSTING MALES

The stag beetle fights to overturn the opposition. The mandibles have a series of prongs that give a good grip as the fighters try to flip their rival. However, injuries are rare.

◁ FREQUENT FLIER

Despite their large size and cumbersome appearance, stag beetles are frequent fliers. The males fly more often than the females, as they patrol their territories.

